Find out the rate of commission on the items A, B and C by using Cramer's rule.

10. The subscription department of a magazine sends out a letter to a large mailing list inviting subscriptions for the magazine. Some of the people receiving this letter already subscribe to the magazine while others do not. From this mailing list, 60% of those who already subscribe will subscribe again while 25% of those who do not now subscribe will subscribe. On the last letter it was found that 40% of those receiving it ordered a subscription. What percent of those receiving the current letter can be expected to order a subscription?

Summary

In this chapter we have acquired the knowledge of:

- **Rank of a matrix**
  The rank of a matrix A is the order of the largest non-zero minor of A
- The rank of a matrix A is the order of the largest non-zero minor of A
- $\rho(A) \geq 0$
- If A is a matrix of order $m \times n$, then $\rho(A) \leq$ minimum of $\{m, n\}$
- The rank of a zero matrix is '0'
- The rank of a non- singular matrix of order $n \times n$ is 'n'
- **Equivalent Matrices**
  Two Matrices A and B are said to be equivalent if one can be obtained from another by a finite number of elementary transformations and we write it as $A \sim B$.
- **Echelon form**
  A matrix of order $m \times n$ is said to be in echelon form if the row having all its entries zero will lie below the row having non-zero entry.
- A system of equations is said to be consistent if it has at least one set of solution. Otherwise, it is said to be inconsistent
  - If $\rho([A, B]) = \rho(A)$, then the equations are consistent.
  - If $\rho([A, B]) = \rho(A) = n$, then the equations are consistent and have unique solution.
  - If $\rho([A, B]) = \rho(A) < n$, then the equations are consistent and have infinitely many solutions.
  - If $\rho([A, B]) \neq \rho(A)$ then the equations are inconsistent and has no solution.
- $|adjA| = |A|^{n-1}$
- If $|A| = 0$ then A is a singular matrix. Otherwise, A is a non singular matrix.
- In $AX = B$ if $|A| \neq 0$ then the system is consistent and it has unique solution.
- Cramer's rule is applicable only when $\Delta \neq 0$.

XII Std - Business Maths & Stat EM Chapter 1.indd 23 2/28/2020 5:28:31 PM